

1. Recognize languages

$$M = (K, \Sigma, \delta, s, H)$$

↓
tape alphabet

input alphabet: $\Sigma_0 \subseteq (\Sigma - \{\triangleright, \sqcup\})$ $M = (K, \Sigma_0, \Sigma, \delta, s, H)$

initial configuration: given any $w \in \Sigma_0^*$,

M starts with $(s, \triangleright \sqcup w)$



$(s, \triangleright \sqcup w) \vdash_n^* (h, \dots)$ for some $h \in H$.

$$L(M) = \{ w \in \Sigma_0^* : M \text{ halts on } w \}$$

M semidecides $L(M)$

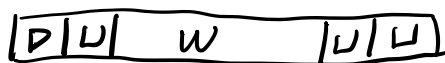
↘ recursively enumerable.

Example.

Fix some input alphabet Σ_0 . Let $a \in \Sigma_0$.

R_a semidecides $L = \{ w \in \Sigma_0^* : w \text{ contains } a \}$

$\rightarrow R \curvearrowright \bar{a}$



yes no
↑ ↑

Let $M = (K, \Sigma, \delta, s, \{y, n\})$ be a TM with input alphabet Σ_0 .

We say M decides a language $L \subseteq \Sigma_0^*$ if

(1) for every $w \in L$, $(s, \triangleright \sqcup w) \vdash_n^* (y, \underbrace{\triangleright u a v}_{\text{accepting}})$ for some $u, v \in \Sigma^*$
 $a \in \Sigma$

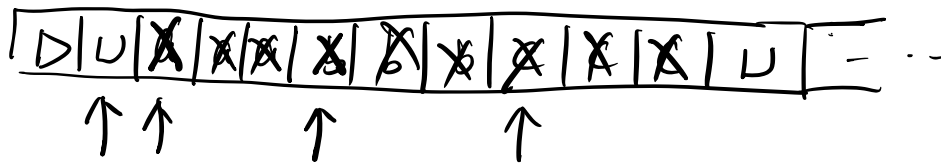
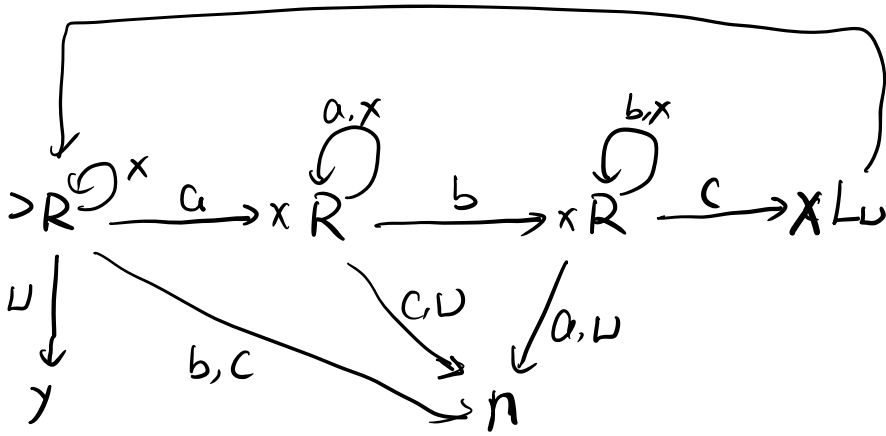
configuration

(

A language is recursive/decidable if it is decided by some TM.

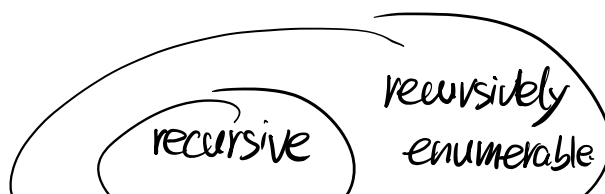
Example

The following TM decides $\{a^n b^n c^n : n \geq 0\}$



Theorem

If L is recursive, it must be recursively enumerable.



Proof:

L is recursive

$\Rightarrow \exists M = (K, \Sigma, \delta, s, \gamma(n))$ decides L

$\downarrow$
looping state

$\delta(n, a) = (n, a)$ for any $a \in \Sigma - \{\square\}$

$\delta(n, \square) = (n, \rightarrow)$

Theorem

If L is recursive, so is $\bar{L}$.

Idea

exchange the role of γ and n

If L is recursively enumerable, so is $\bar{L}$? False

Compute function

Let $M = (K, \Sigma, \delta, s, H)$ be a TM with input alphabet Σ_0 .

Let $w \in \Sigma_0^*$.

If $(s, \square \sqcup w) \vdash_M^* (h, \square \sqcup y)$ for some $h \in H$ and $y \in \Sigma_0^*$.

y : the output of M on w , $M(w)$

$\square$	$\sqcup$	$y \in \Sigma_0^*$
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Let $f: \Sigma_0^* \rightarrow \Sigma_0^*$. We say M computes f if
 for any $w \in \Sigma_0^*$, $M(w) = f(w)$.

(M halts with $\triangleright \sqcup f(w)$)

A function is recursive/computable if some TM computes it.

Example

$f(w) = ww$ is computed by the following TM.

$\triangleright C R \sqcup^2 S \leftarrow L \sqcup$

$\downarrow$
copy machine

$\triangleright \sqcup w$

$\triangleright \sqcup w \sqcup w \sqcup$

$\triangleright \sqcup w \sqcup w \sqcup$

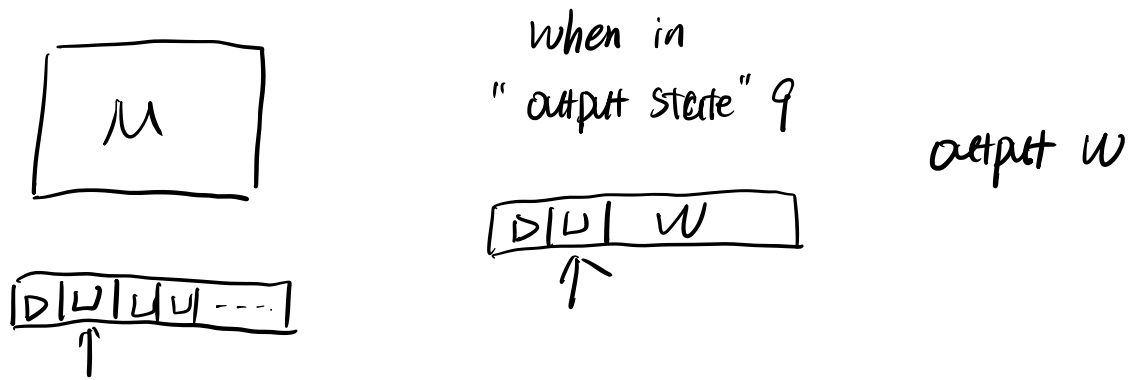
$\triangleright \sqcup w w \sqcup$

$\triangleright \sqcup w w \sqcup$

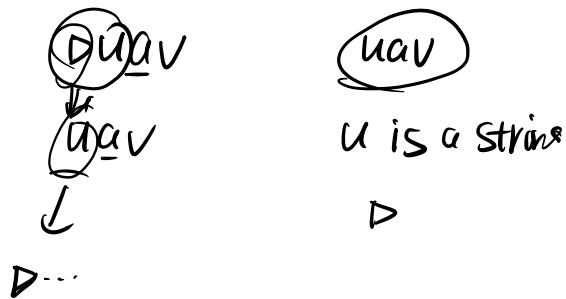
Enumerator

We say a TM M enumerates a language L
 if for some state q ,

$$L = \{ w : (S, \underline{\Delta}) \vdash_m^* (q, \underline{\Delta} w) \}$$

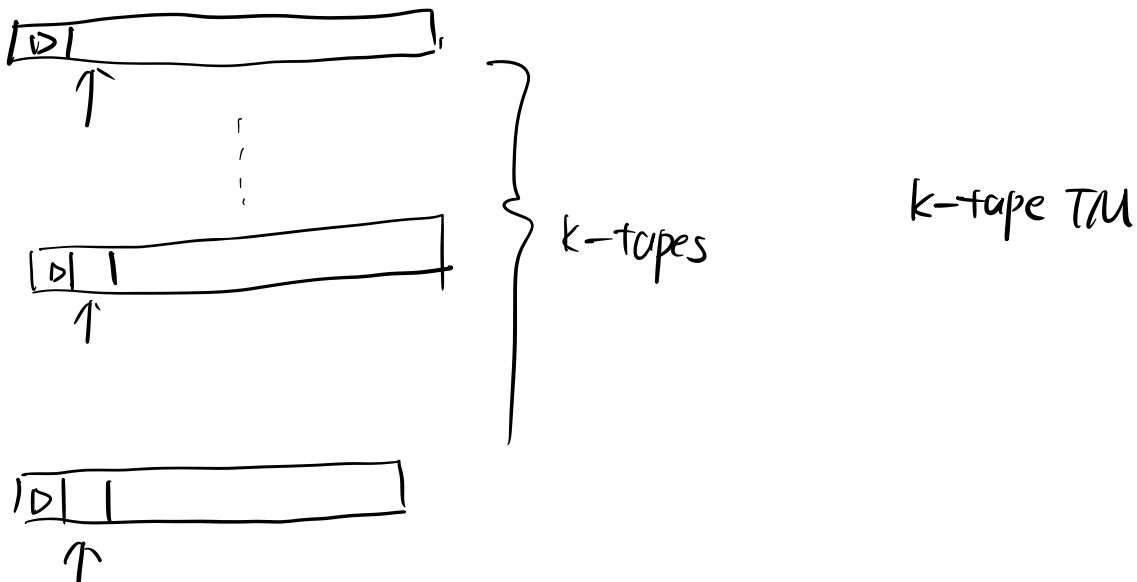


A language is Turing enumerable if some TM enumerates it.



Extension of TM

1. Multiple tape.

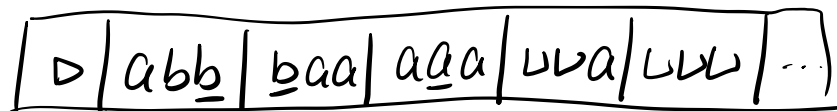
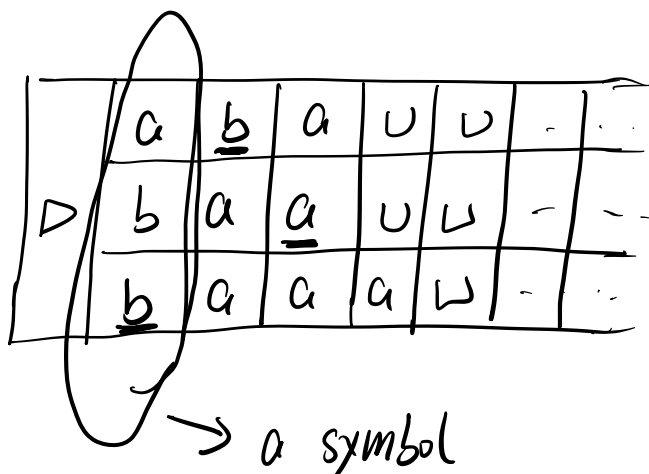
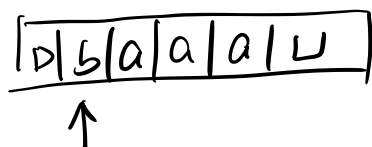
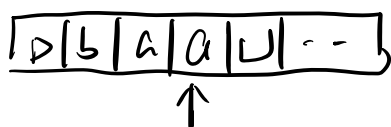
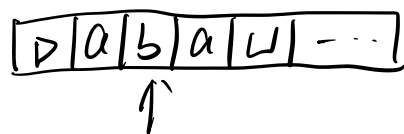


$$\delta: (k-1) \times \Sigma^k \rightarrow K \times (\Sigma \cup \{\leftarrow, \rightarrow\})^k$$

Theorem

Any k -tape Turing machine can be simulated by a standard TM.

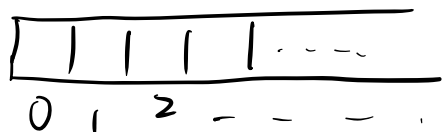
k -track (conceptually)



$$\Sigma' = (\Sigma \cup \{a = a \in \Sigma\})^k$$

2. Two-way infinite tape

infinite

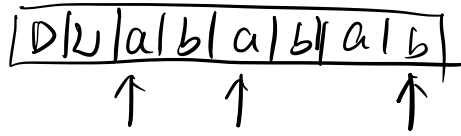


2-tape TM

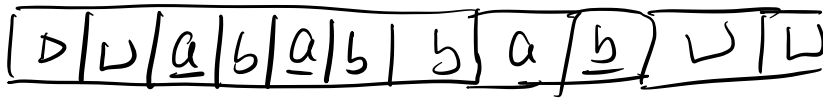


-1 -2 -3

3. Multiple head

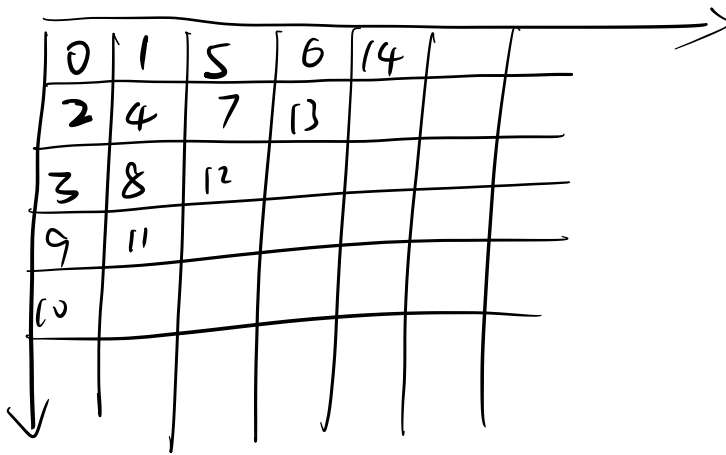


k-head TM.

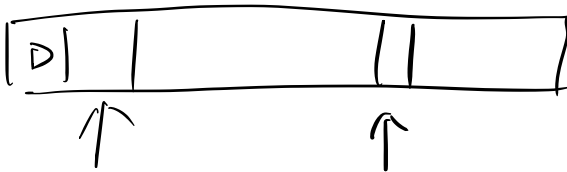


standard TM

4. Two dimensional tape



5. random access



6. Non-determinism

deterministic TM (DTM)

A non-deterministic Turing machine (NTM) is 5-tuple $(K, \Sigma, \Delta, S, H)$

- K : - - -
- Σ : - - -
- S : - - -
- H : - - -
- $\Delta \subseteq ((K-H) \times \Sigma) \times (K \times (\Sigma \cup \{\leftarrow, \rightarrow\}))$ relation

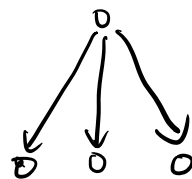
configuration (q, dabaab)

t_M	t_M^*	t_M^N
one step	several steps	N step

DTM



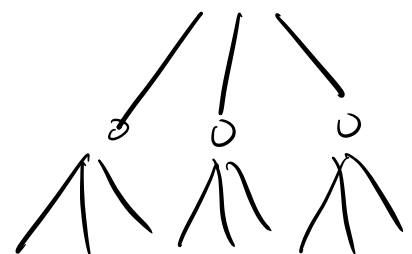
NTM



(S, daab)



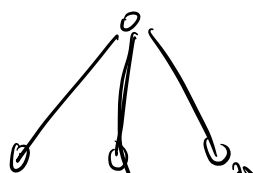
(S, daab)

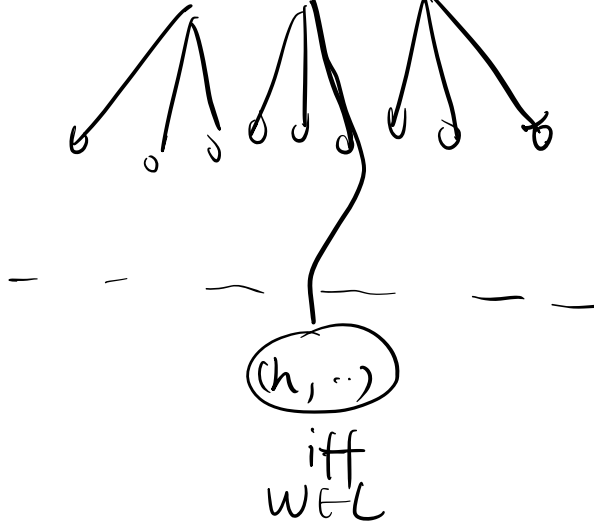


A NTM N with input alphabet Σ_0 semidecides a language $L \subseteq \Sigma_0^*$ if for any $w \in \Sigma_0^*$, the following is true.

$w \in L$ iff $(S, \text{daab}) \xrightarrow{t_M^*} (h, \text{daab})$ for some h .

given w



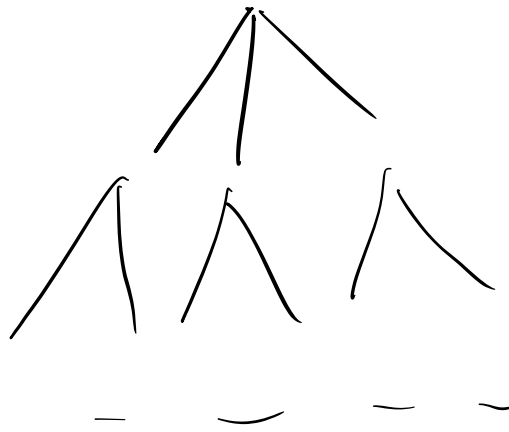


Let $M = (K, \Sigma, \Delta, s, \gamma, n)$ be a NTM with input alphabet Σ .
 We say M decides a language $L \subseteq \Sigma^*$ if the followings are true.

- (1) there is a natural number N , depending on M and w ,
 such that there is no configuration C satisfying

$$(s, \triangleright \sqcup w) \vdash_M^N C$$

Given w



every branch halts within $N-1$ steps.

- (b) $w \in L$ if and only if $(s, \triangleright \sqcup w) \vdash_M^* (\gamma, \triangleright \sqcup \alpha \sqcup v)$

Example

Let C be the set of binary strings representing composite number

30 100 100 100 100

Design a NTM decides C .



(non-deter.) Generate
a binary string
 $2 \leq |p| \leq |w|$

$$2 \leq |g| \leq |v|.$$

1. multiply p with q , compare product with w
2. if match
 halt with y
else
 halt with n

Theorem

A NTM N can be simulated by a DTM M .

Proof:

 N semidecides L DTM M semidecides L

given w

 $(S, \underline{DLW})$

1

 γ r^2 

$(h, \Delta \dots)$

iff
 $w \in L$

DFS bad

BFS good

$$\Delta \subseteq (K-H) \times \Sigma \times (K \times (\Sigma \cup \{\leftarrow, \rightarrow\}))$$

$$|K| \times (|\Sigma| + 2) = r$$

$$\text{fanout} \leq r$$

3-tape DTM to simulate NTM

$\boxed{D|U|W}$ input

$\boxed{D|U|W}$ simulate a branch of NTM

$\boxed{D|124}$ enumerate strings over
 $\{1, 2, \dots, r\}$

$1, 2, 3, \dots, r, 11, 12, \dots, 1r$
 $21, \dots, 2r$

